

# Aayush Sangani

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## EDUCATION

### University of California, San Diego

La Jolla, CA

*Bachelors of Science in Computer Engineering*

*Expected June 2027*

- **Relevant Coursework:** Computer Architecture, Digital Logic Design, ASIC and FPGA Design, Microarchitecture, Advanced VLSI, Systems Programming, Data Structures, Linear Algebra

## TECHNICAL SKILLS

**Hardware & HDL:** SystemVerilog, Verilog, ARM/RISC-V Assembly, x86 Assembly, RTL Design, Physical Design

**EDA/CAD Tools:** Synopsys Fusion Compiler, PrimeTime, Cadence Innovus, Tempus, QuestaSim, Quartus

**Languages:** C/C++, Python, TCL, Perl, Bash/Shell, TypeScript, JavaScript, SQL, Java, HTML/CSS

**Frameworks & Tools:** PyTorch, scikit-learn, pandas, React, FastAPI, Docker, AWS, PostgreSQL, Linux, Git

## EXPERIENCE

### Intel

Folsom, CA

*Hardware Engineering Intern – CPU Physical Design*

*June 2026 – September 2026*

- Performed **cluster/full-chip STA** in **PrimeTime**, debugging **setup/hold violations** across **PVT corners** on smashed-macro, hard-IP, and I/O constrained paths via physical placement, clock adjustments, and RTL escalation
- Built an **MCP AI agent** using custom **Tcl/Python** automation and Markdown skills to parse live **PrimeTime** paths, identify **setup/hold** violations, and recommend timing fixes, reducing cluster/full-chip analysis by **50%**
- Executed cluster-level **RTL-to-GDS** physical design for **SMT-enabled P-core blocks** in **Synopsys Fusion Compiler**, driving **floorplanning**, macro placement, **PnR**, **CTS**, and timing/congestion closure
- Leveraged **Cadence Cerebrus** for Agentic AI driven design exploration and implementation-flow optimization, analyzing **QoRs** across CPU cluster stages to improve **PPA**, and timing convergence

### California Franchise Tax Board

Sacramento, CA

*Software Engineering Intern*

*January 2025 – September 2025*

- Automated disk audits via **PowerShell**, executing **SQL Server** 2016–2022 migrations across **10+** environments
- Developed **PowerShell** and **SQL** scripts to migrate logins and identify linked server dependencies
- Configured service accounts, permissions, and firewall rules with Security and File-app teams for **SQL Server** 2022
- Maintained parallel fallback servers for **1–2** months, completing final cutover after validating all dependencies

### Key Business Solutions

Sacramento, CA

*Software Developer Intern*

*July 2024 – August 2025*

- Diagnosed and resolved **React** state mutation bugs across **15+** chatbots, eliminating recurring UI defects
- Audited **API** call chains via **Chrome DevTools** and triaged client-reported bugs across **15+** chatbots via **Jira**
- Deployed frontend builds on **AWS EC2**, isolating failures via **PM2** logs; reducing resolution under **30 minutes**

## PROJECTS

### Argus – Parameterized N-bit LFSR BIST Architecture

- Built a self-testing hardware block to reduce the need for expensive external **ATE**, embedding on-chip pattern generation and output compaction to reduce silicon test time and improve **fault coverage** at **tapeout**
- Designed a parameterized N-bit **LFSR** in synthesizable **SystemVerilog** RTL as a **PRPG**, generating  $2^N - 1$  deterministic test patterns through configurable **tap logic**, **seed initialization**, and **XOR feedback**
- Extended to a full **BIST** architecture with **MISR** output compactor and **FSM** test controller, validating **stuck-at fault** detection through injection in **QuestaSim** with **SVA** assertions and **waveform analysis**
- Synthesized in **Quartus** across  $N=4$  and  $N=8$ , analyzing **LUT/FF** utilization, **critical path** timing, and **Fmax** to evaluate **PPA** tradeoffs across **FPGA** and **ASIC**-oriented physical design parameter sweeps

### Midas – Financial News Intelligence Platform

- Built a **FastAPI** pipeline scoring news and **SEC EDGAR** filings into **BUY/HOLD/SELL** signals per ticker
- Deployed **FinBERT** via **PyTorch**, persisting **time-weighted** scores to **PostgreSQL** on an **APScheduler** job
- Ingested **Yahoo/Google RSS** and **Finnhub** news feeds, deduplicating and cleaning articles before inference
- Containerized with **Docker**, exposing a **REST API** and **Python CLI** over the ingestion pipeline